Can a metric be reconstructed from a connection?

For a given connection $\Gamma^\lambda_{\mu\nu}$, can we derive the metric $g_{\mu\nu}$ standing behind it?

Short answer

Sometimes, but not for an arbitrary connection. If Γ is the Levi-Civita connection of a metric, then the metric must satisfy

$$\nabla_\lambda g_{\mu\nu}=0,$$

or, in coordinates,

$$\boxed{ \partial_\lambda g_{\mu\nu} =\Gamma^\rho_{\lambda\mu}g_{\rho\nu} + \Gamma^\rho_{\lambda\nu}g_{\mu\rho}. }$$

This is a linear, first-order system of partial differential equations for the components of g. A solution must also be symmetric, non-degenerate, and have the desired signature. In addition, a Levi-Civita connection must be torsion-free:

$$\Gamma^\lambda_{\mu\nu} =\Gamma^\lambda_{\nu\mu}.$$

Thus, knowing a connection does not guarantee that a metric exists. When a metric does exist, the connection usually determines it only up to an overall nonzero constant:

$$g_{\mu\nu}\longmapsto Cg_{\mu\nu},$$

because a constant rescaling does not change the Christoffel symbols.

Existence and integrability

Choose a symmetric matrix $g_{\mu\nu}(p)$ at one point p. The equation $\nabla g=0$ parallel-transports this matrix to nearby points. The result is independent of the path precisely

when the chosen matrix is preserved by the holonomy of the connection. For every closed loop based at pp, with parallel-transport matrix PP, one needs

PTg(p)P=g(p). P^T g(p)P=g(p).

A useful local necessary condition follows by commuting two covariant derivatives:

0=[∇α,∇β]gμν=−Rρμαβgρν−Rρναβgμρ. 0=[\nabla_\alpha, \nabla_\beta]g_{\mu\nu} =-R^\rho{}_{\mu\alpha\beta} g_{\rho\nu} -R^\rho{}_{\nu\alpha\beta}g_{\mu\rho}.

At a point, these are linear algebraic equations for the unknown entries of gμν(p)g_{\mu\nu}(p). Curvature transported from other points, or equivalently the full holonomy condition, supplies the complete integrability test. Consequently, a practical reconstruction procedure is:

1. Check that the connection is torsion-free.
2. Solve the curvature/holonomy constraints for a symmetric, non-degenerate g(p)g(p) of the desired signature.
3. Integrate ∇λgμν=0\nabla_\lambda g_{\mu\nu}=0 away from pp.
4. Substitute the result into the Christoffel formula and verify that it reproduces the original connection.

The solution need not always be unique up to one constant. The set of all parallel symmetric bilinear forms is a vector space. Reducible or flat connections can preserve several independent metrics. For a generic irreducible metric holonomy, however, the only freedom is the common constant scale.

If the supplied connection has torsion, it cannot be the Levi-Civita connection of any metric. It could still be compatible with a metric in a theory with torsion, but then metric compatibility alone generally does not select a unique metric.

Test: the Schwarzschild connection

Work in Schwarzschild coordinates (t,r,θ,ϕ)(t,r,\theta,\phi), away from the coordinate singularity at r=2Mr=2M, and put

f(r)=1−2Mr. f(r)=1-\frac{2M}{r}.

Suppose that only the Schwarzschild connection is given. Some of its nonzero coefficients are

Γttr=Γtrt=f′2f,Γrtt=ff′2,Γrrr=−f′2f,Γrθθ=−rf,Γrϕϕ=−rfsin2θ,Γθrθ=Γϕrϕ=1r,Γθϕϕ

To try to reconstruct the starting metric, use a diagonal, static, spherically symmetric ansatz in these coordinates:

ds2=−A(r)dt2+B(r)dr2+C(r)dθ2+D(r)sin2θdϕ2. ds^2=-A(r)dt^2+B(r)dr^2+C(r)d\theta^2 +D(r)\sin^2\theta\,d\phi^2.

The radial metric-compatibility equations give

A′A=2Γtrt=f′f,B′B=2Γrrr=−f′f, \frac{A'}{A}=2\Gamma^t{}_{rt} =\frac{f'}{f}, \qquad \frac{B'}{B}=2\Gamma^r{}_{rr}=-\frac{f'}{f},

and

C′C=2Γθrθ=2r,D′D=2Γϕrϕ=2r. \frac{C'}{C}=2\Gamma^\theta{}_{r\theta}=\frac{2}{r}, \qquad \frac{D'}{D}=2\Gamma^\phi{}_{r\phi}=\frac{2}{r}.

Therefore,

A=ctf,B=crf,C=cθr2,D=cϕr2. A=c_t f,\qquad B=\frac{c_r}{f}, \qquad C=c_\theta r^2,\qquad D=c_\phi r^2.

The angular compatibility equation with indices (λ,μ,ν)=(ϕ,θ,ϕ) (\lambda,\mu,\nu)=(\phi,\theta,\phi) gives

0=Γϕϕθgϕϕ+Γθϕϕgθθ, 0=\Gamma^\phi{}_{\phi\theta} g_{\phi\phi} +\Gamma^\theta{}_{\phi\phi}g_{\theta\theta},

so cϕ=cθc_\phi=c_\theta. The equations for components that vanish in the diagonal ansatz are also important. For example,

0=∇tgtr=−Γrttgrr−Γttrgtt 0=\nabla_t g_{tr} =-\Gamma^r{}_{tt} g_{rr}-\Gamma^t{}_{tr}g_{tt}

implies cr=ctc_r=c_t, while

0=∇θgrθ=−Γθθrgθθ−Γrθθgrr 0=\nabla_\theta g_{r\theta} =-\Gamma^\theta{}_{\theta r}g_{\theta\theta} -\Gamma^r{}_{\theta\theta}g_{rr}

implies cθ=crc_\theta=c_r. Hence all four constants coincide:

ct=cr=cθ=cϕ=c. c_t=c_r=c_\theta=c_\phi=c.

The reconstructed metric is therefore

ds2=c[−(1−2Mr)dt2+(1−2Mr)−1dr2+r2dθ2+r2sin2θdϕ2]. \boxed{ ds^2=c\left[-\left(1-\frac{2M}{r}\right)dt^2 +\left(1-\frac{2M}{r}\right)^{-1}dr^2 +r^2d\theta^2+r^2\sin^2\theta\,d\phi^2 \right]. }

For c>0c>0 this has the usual (−,+,+,+)(-,+,+,+) signature. Thus the Schwarzschild connection reconstructs the Schwarzschild metric, with exactly the expected undetermined overall constant. The mass parameter MM is already encoded in the connection and is recovered in the radial function f(r)f(r).

The M=0M=0 case deserves one qualification: the connection is then flat. Without imposing the static, diagonal, spherically symmetric ansatz, a flat connection preserves a larger family of constant bilinear forms in affine Cartesian coordinates. This is an example in which the same connection can come from more than constant rescalings of one metric.

Test: the Taub-NUT connection

The Lorentzian Taub-NUT metric has an off-diagonal tt–ϕ\phi component, so it provides a more substantial test. We use the convention

Σ(r)=r2+ℓ2,Δ(r)=r2−2Mr−ℓ2,F(r)=ΔΣ, \Sigma(r)=r^2+\ell^2, \qquad \Delta(r)=r^2-2Mr-\ell^2, \qquad F(r)=\frac{\Delta}{\Sigma},

and define

q(θ)=2ℓcosθ,η=dt+q(θ)dϕ. q(\theta)=2\ell\cos\theta, \qquad \eta=dt+q(\theta)d\phi.

The usual metric in this convention is

ds2=−Fη2+dr2F+Σ(dθ2+sin2θdϕ2). ds^2=-F\eta^2+\frac{dr^2}{F} +\Sigma\left(d\theta^2+\sin^2\theta\,d\phi^2\right).

Changing the convention from dt+2ℓcosθdϕdt+2\ell\cos\theta\,d\phi to dt−2ℓcosθdϕdt-2\ell\cos\theta\,d\phi is equivalent to changing the sign of ℓ\ell in the formulas below.

Reading the structure from the connection

Some connection coefficients that are sufficient for the reconstruction are

Γtrt=F′2F,Γϕrt=0,Γrtt=FF′2,Γrrr=−F′2F,Γθrθ=Γϕrϕ=rΣ,Γrθθ=−rF,Γtrϕ=q2(F′F−

As usual, coefficients related by symmetry of the two lower indices are understood. The characteristic Taub-NUT twist can be read directly from the connection:

ΓrtϕΓrtt=q(θ)=2ℓcosθ, \frac{\Gamma^r{}_{t\phi}}{\Gamma^r{}_{tt}} =q(\theta)=2\ell\cos\theta,

where the denominator is nonzero. Similarly,

F=−1rΓrθθ,Σ=rΓθrθ. F=-\frac{1}{r}\Gamma^r{}_{\theta\theta}, \qquad \Sigma=\frac{r}{\Gamma^\theta{}_{r\theta}}.

Thus the radial functions and the twisted one-form are encoded in the connection rather than being additional metric data.

Solving the compatibility equations

Guided by these symmetries, take the ansatz

ds2=−A(r)η2+B(r)dr2+C(r)(dθ2+sin2θdϕ2). ds^2=-A(r) \eta^2+B(r)dr^2 +C(r)\left(d\theta^2+ \sin^2\theta\,d\phi^2\right).

In the coordinate basis its nonzero components are

gtt=−A,gtϕ=−Aq,grr=B,gθθ=C,gϕϕ=Csin2θ−Aq2. \begin{aligned} g_{tt}&=-A, &g_{t\phi}&=-Aq, &g_{rr}&=B,\\ g_{\theta\theta}&=C, &g_{\phi\phi}&=C\sin^2\theta-Aq^2. \end{aligned}

The (r,t,t)(r,t,t), (r,r,r)(r,r,r), and (r,θ,θ)(r,\theta,\theta) components of ∇g=0\nabla g=0 give

A′A=2Γtrt=F′F,B′B=2Γrrr=−F′F, \frac{A'}{A}=2\Gamma^t{}_{rt}=\frac{F'}{F}, \qquad \frac{B'}{B}=2\Gamma^r{}_{rr}=-\frac{F'}{F},

and

C′C=2Γθrθ=2rΣ. \frac{C'}{C}=2\Gamma^\theta{}_{r\theta} =\frac{2r}{\Sigma}.

After integration,

A=aF,B=bF,C=cΣ, A=aF, \qquad B=\frac{b}{F}, \qquad C=c\Sigma,

where a,b,ca,b,c are initially independent constants. As in the Schwarzschild example, equations for components that vanish in the ansatz relate these constants. First,

0=∇tgtr=−Γrttgrr−Γttrgtt−Γϕtrgtϕ=F′2(a−b), \begin{aligned} 0=\nabla_t g_{tr} &=-\Gamma^r{}_{tt}g_{rr} -\Gamma^t{}_{tr}g_{tt} -\Gamma^\phi{}_{tr}g_{t\phi}\\ &=\frac{F'}{2}(a-b), \end{aligned}

so a=ba=b. Next,

0=∇θgrθ=−Γθθrgθθ−Γrθθgrr=r(b−c), \begin{aligned} 0=\nabla_\theta g_{r\theta} &=-\Gamma^\theta{}_{\theta r} g_{\theta\theta} -\Gamma^r{}_{\theta\theta}g_{rr}\\ &=r(b-c), \end{aligned}

so b=cb=c. Hence

a=b=c=k. a=b=c=k.

The compatibility equations for gtϕg_{t\phi} and gϕϕg_{\phi\phi} are then automatically satisfied. For instance, the (r,t,ϕ)(r,t,\phi) equation uses both Γtrϕ\Gamma^t{}_{r\phi} and Γϕrϕ\Gamma^\phi{}_{r\phi} and returns ∂rgtϕ=−qA′\partial_r g_{t\phi}=-qA', as required.

The reconstructed metric is therefore

ds2=k[−r2−2Mr−ℓ2r2+ℓ2(dt+2ℓcosθdϕ)2+r2+ℓ2r2−2Mr−ℓ2dr2+(r2+ℓ2)(dθ2+sin2θdϕ2)]. \boxed{ \begin{aligned} ds^2=k\Bigg[& -\frac{r^2-2Mr-\ell^2}{r^2+\ell^2} \left(dt+2\ell\cos\theta\,d\phi\right)^2\\ &+\frac{r^2+\ell^2}{r^2-2Mr-\ell^2}\,dr^2 +(r^2+\ell^2) \left(d\theta^2+ \sin^2\theta\,d\phi^2\right) \Bigg]. \end{aligned} }

For k>0k>0, this is Lorentzian wherever F≠0F\neq0; the coordinate vector ∂t\partial_t is timelike in a region where F>0F>0. Therefore, on a generic non-flat region, the Taub-NUT connection recovers the starting Taub-NUT metric up to the same unavoidable overall scale as in the Schwarzschild case. Both MM and the NUT parameter ℓ\ell occur in the connection.

The limit ℓ→0\ell\to0 gives

η→dt,Σ→r2,F→1−2Mr, \eta\to dt, \qquad \Sigma\to r^2, \qquad F\to1-\frac{2M}{r},

so the reconstruction reduces continuously to the Schwarzschild result. This also checks that the off-diagonal terms have the correct normalization and sign.

This calculation is local. Globally, the one-form η\eta requires the usual Taub-NUT patching around the polar axes, and questions involving the Misner string or a periodic time coordinate require global information that cannot be inferred from connection coefficients in a single coordinate chart.

Conclusion

A connection determines a metric exactly when it is torsion-free and its holonomy preserves a non-degenerate symmetric bilinear form of the required signature. Reconstruction means solving ∇g=0\nabla g=0. For the generic Schwarzschild and Taub-NUT connections, this procedure returns the original metric up to a constant overall scale, which no Levi-Civita connection can determine. The Taub-NUT example additionally shows that off-diagonal metric components can be reconstructed: their twisting function is already encoded in the mixed connection coefficients.